A novel framework to study configural and holistic processing

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Background

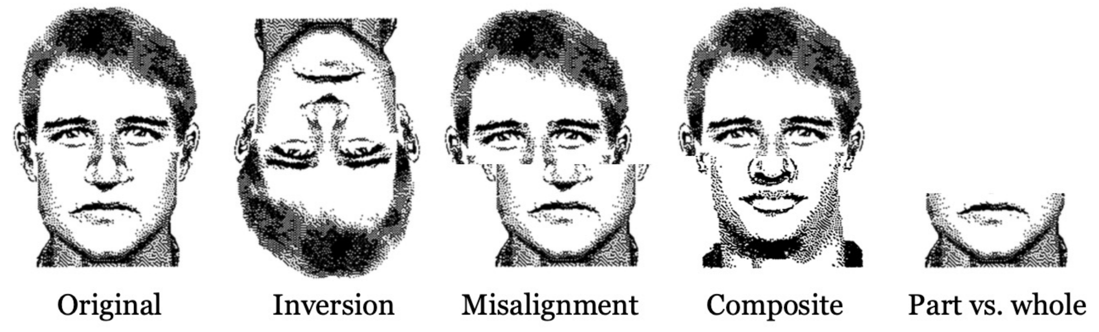
The integral role of holistic processing in face recognition, which sets it apart from other visual processing techniques, is well-established[1]. As such, phenomena associated with holistic processing are often recognized as robust markers for face perception[2].

However, inconsistencies arise when examining holistic processing within the context of object visual perception [3,4,5], leaving the underlying cause of holistic processing and its potential limitations within face perception an open-ended question.

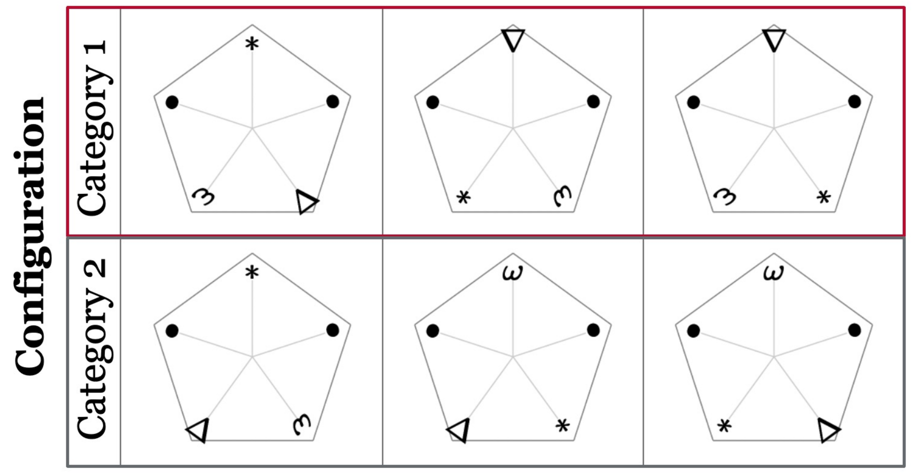
In this study, we explore the correlation between configuration and holistic processing, utilizing both human behavioral data and convolutional neural networks (CNNs). Our aim is to determine if the presence of pure configural information alone is sufficient to initiate holistic processing.

Methods

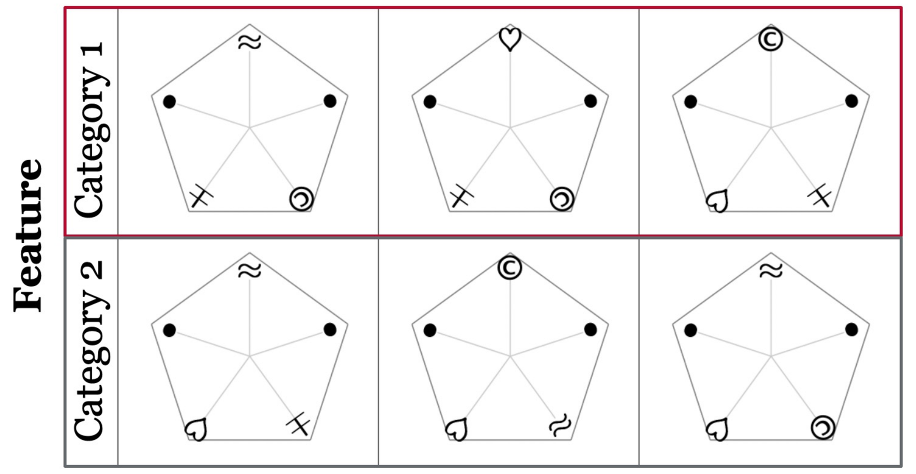
In order to assess holistic processing, we explored four widely recognized effects associated with it: *inversion*, *misalignment*, *composite*, and *part vs. whole* effect.



Adapted from [6,7]



Only based on *relative location*, features are identical



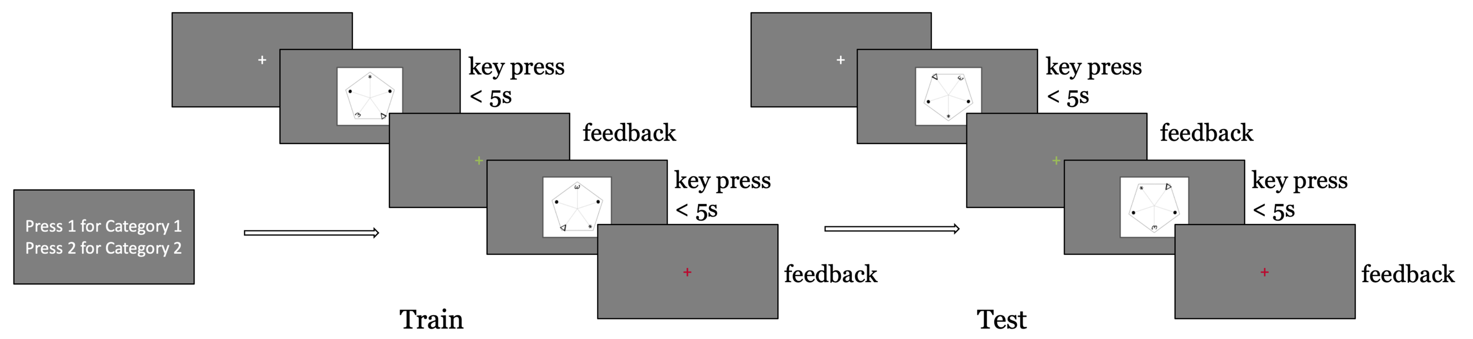
Only based on *paired features*, disregard relative location

Stimuli:

The stimuli used for *configuration* were randomly assigned to two categories. These stimuli differ solely in configuration but are composed of identical features.

As a control measure, *featural* stimuli are classified by which features are present. Each category contains 2 target features, with 1 feature from the other category (e.g. Category 1: "±", "©" vs. Category 2: "♡", "≈"). The relative locations of these features did not inform category membership.

Behavioral task:



CNNs:

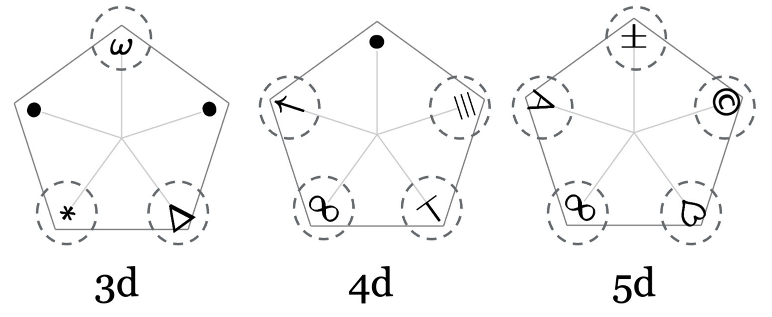
Initialized by the pre-trained ResNet-50 models on ImageNet, and trained on the above stimuli with random noise added to increase size of the training set.

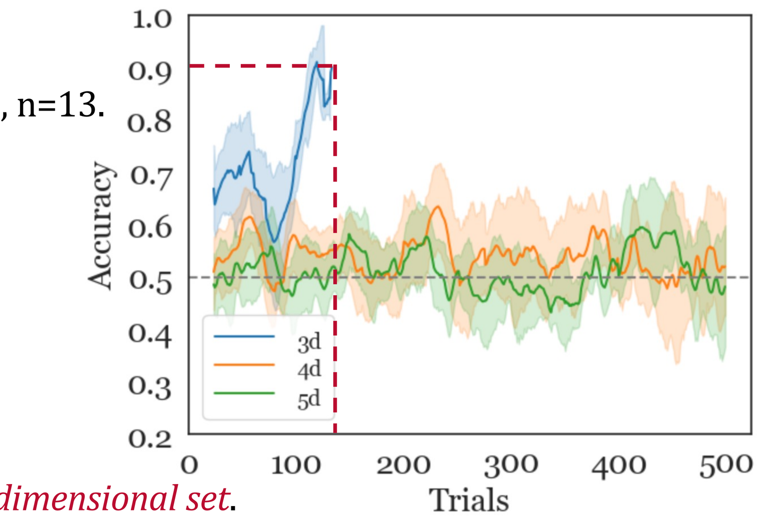
Trained two groups (each n=15) of models for configuration and feature (to 95% acc. of the trained stimuli).

Exp0: Can humans learn configuration-based categories?

And how much dimensions can they learn?

We used *3*, *4*, and *5 dimensional stimuli* (only for the training session), n=13.





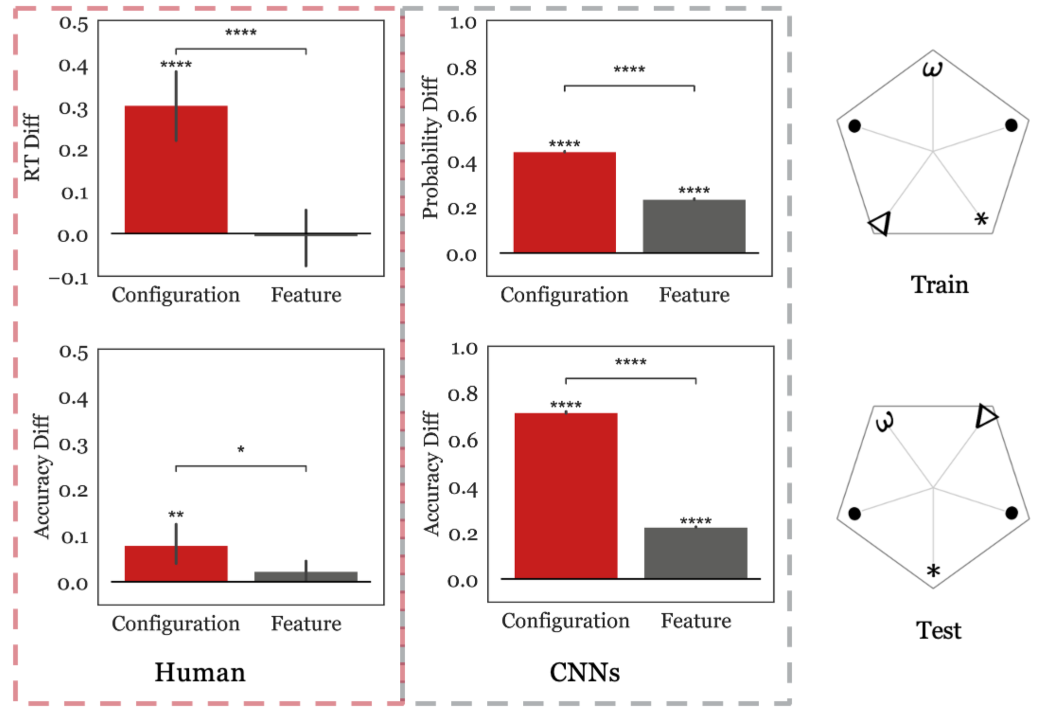
Results: humans *can* learn to categorize configural stimuli, for the *3 dimensional set*.

Exp1: Inversion effect

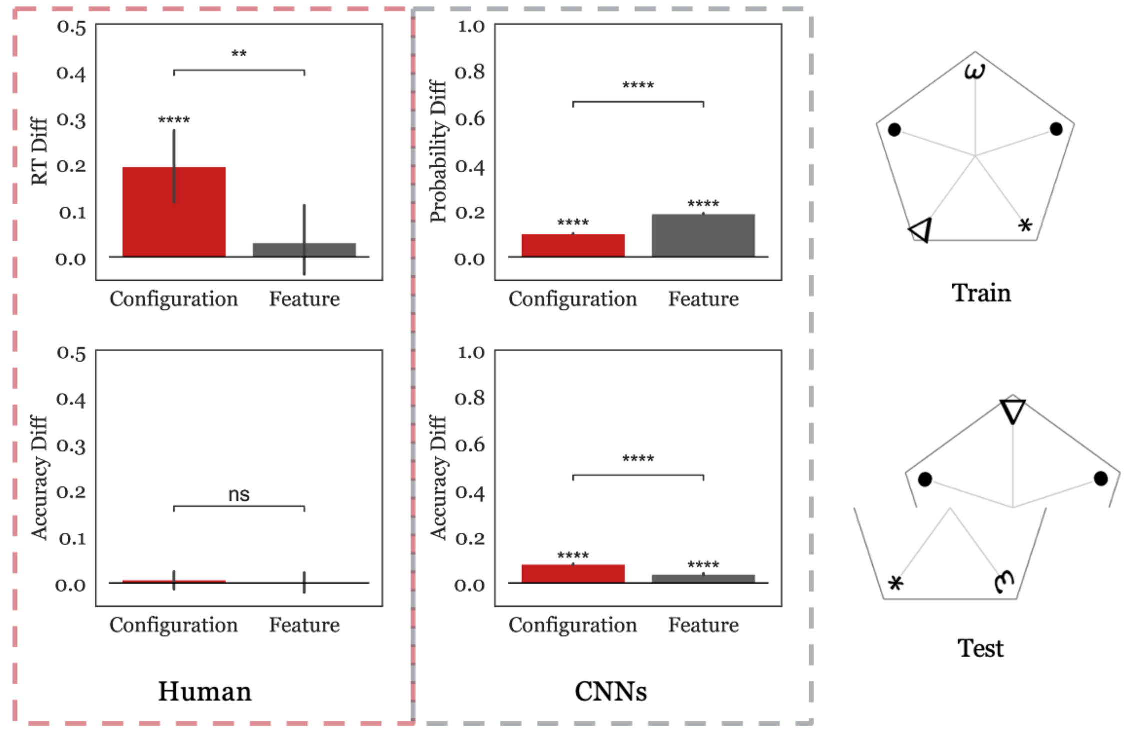
We recruited *15 human participants* who underwent training in both configuration and feature conditions.

Similarly, *15 models* were trained for each respective condition.

The results from both humans and models displayed a striking resemblance: *configuration* was found to *have a significant inversion effect* when compared to the feature condition. This finding was supported by tests conducted on reaction time (RT) and accuracy differences.



Exp2: Misalignment effect



This experiment used the same *15* human participants and *15* models from experiment 1.

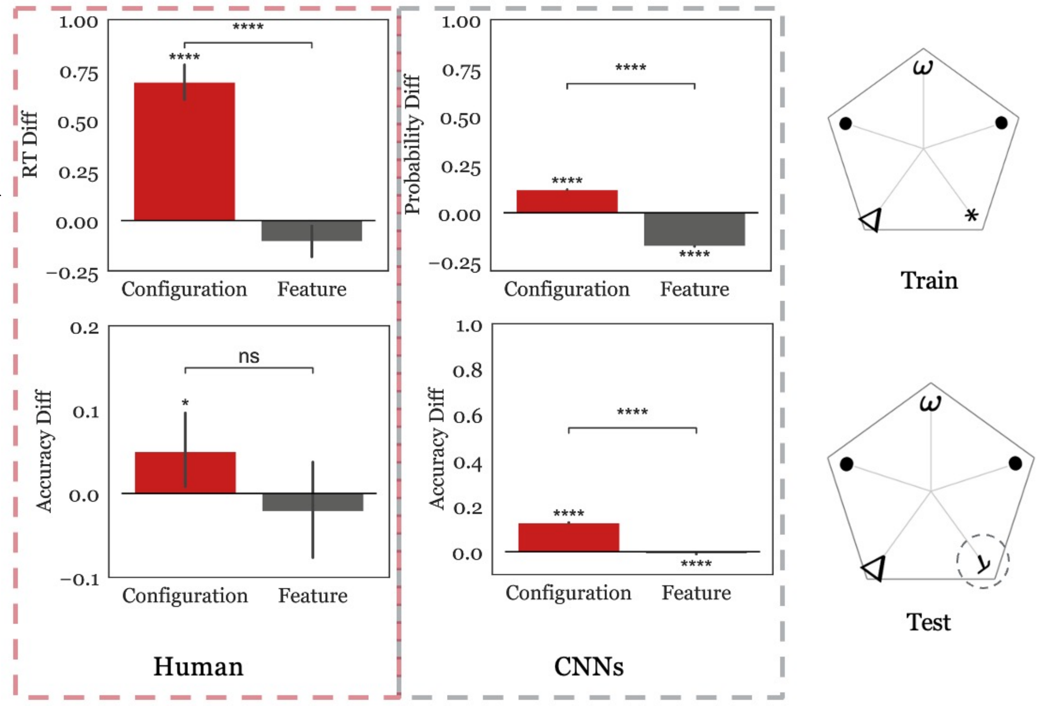
Human participants showed significant RT differences between conditions, demonstrating *a misalignment effect.* We did not observe this effect with accuracy.

The models were more accurate for the original than the misaligned stimuli, especially when they are categorized by configuration than by feature. However, the probability differences were higher for featural stimuli than configural stimuli.

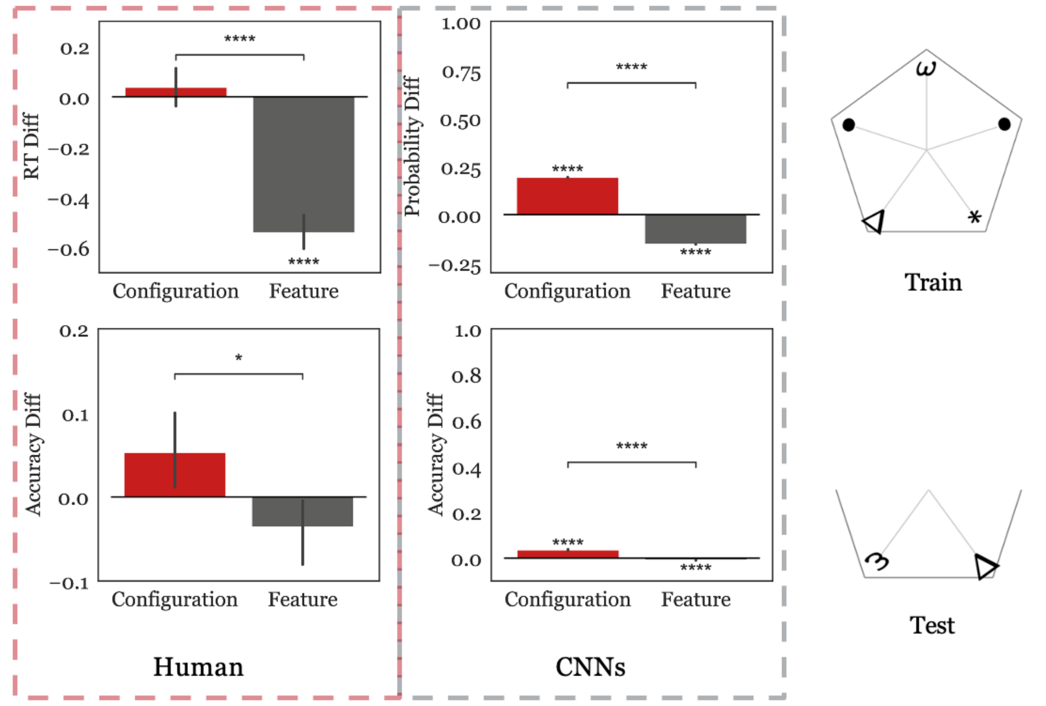
Exp3: Composite effect

We recruited *17 human participants* for training in both configuration and feature conditions. The same *15 models*, trained previously, were employed for each condition.

Similar to the inversion effect findings, both human responses and model outputs consistently demonstrated *a significant composite effect under the configuration condition*. We observed a significant accuracy effect for configural stimuli, and *not* for featural stimuli, although the conditions were not significantly different from each other.



Exp4: Part vs. whole effect



The same set of *17* participants and *15* models from experiment 3 were used.

Consistent findings were observed in both human behavior and model outcomes, all indicating *a strong part vs. whole effect in the configuration condition*, but not in the feature condition.

Conclusions

- We created a unique *set of stimuli* that are *not face-like* but *carry pure configuration information*, in addition to a control stimuli set. This serves as a pioneering framework for the exploration of configural and holistic processing.
- We have demonstrated that *configuration, in isolation, can indeed reproduce effects of holistic processing*, a phenomenon thought to be exclusive to face perception (which is likely the only *natural* stimulus of this kind).
- We are currently investigating the *brain's representation of configuration*, in order to identify the relationship between configural processing and representations of other high-level visual categories, including faces.

References

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